

## SOFTWARE ENGINEERING LAB – 2

(UE24CS341A)

### Agile Backlog Creation & Sprint Simulation in Jira

**Team Number:** 007

**Individual Submission:** Kaustubh Akash (PES1UG24AM131)

**Problem Statement 7:** Hostel Maintenance & Issue Ticketing System

**EPIC:** Epic 3 – Maintenance Work Execution & Progress Tracking (FR-003)

**Note:** This report covers only Epic 3, my individual contribution from the Lab 1 report. The remaining requirements are covered separately by my teammates: Epic 1 – FR-001 & FR-005: P. Saanvi; Epic 2 – FR-002 & NFR-001: Sonia Sharma; Epic 4 – FR-004 & NFR-002: Kasissnu Ssinha.

### Lab Objective

This lab converts the requirements FR-003 from the Lab 1 report into an Agile Epic and User Stories in Jira, estimates effort using the Fibonacci scale, prioritises the backlog, simulates two one-week sprints on a Scrum board, and tracks sprint progress using the Burndown Chart.

### 1. Epic (Derived from FR-003)

The requirement FR-003 forms the backbone of Epic 3: it mandates that maintenance staff update ticket statuses (e.g., In Progress, Resolved) with verified timestamps, ensuring transparency, operational accountability, and accurate maintenance audit logs.

| Epic Key | Epic Name                                      | Mapped Requirements | Description                                                                                                                          |
|----------|------------------------------------------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| HMT-14   | Maintenance Work Execution & Progress Tracking | FR-003              | Provide maintenance technicians with mobile task views, real-time lifecycle status transitions, parts logging, and audit timestamps. |

### 2. User Stories for Epic 3

The Epic is broken into 5 User Stories using the format: As a [role], I want [goal], so that [benefit].

| Key    | Story                            | As a...           | I want...                                                                           | So that...                                                 | Covers |
|--------|----------------------------------|-------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------|--------|
| HMT-15 | 3.1 Assigned Work Queue          | Maintenance Staff | to view a clear mobile list of tickets assigned to me sorted by priority            | I know which room and issue to attend next                 | FR-003 |
| HMT-16 | 3.2 Status Update to In-Progress | Maintenance Staff | to mark a ticket as 'In Progress' when physical work begins                         | the resident and warden know repairs are actively underway | FR-003 |
| HMT-17 | 3.3 Status Update to Resolved    | Maintenance Staff | to transition tickets to 'Resolved' with an automated timestamp upon finishing work | the resident is invited to inspect and sign off            | FR-003 |

|        |                                 |                   |                                                                                 |                                                     |        |
|--------|---------------------------------|-------------------|---------------------------------------------------------------------------------|-----------------------------------------------------|--------|
| HMT-18 | 3.4 Parts Used & Work Log Notes | Maintenance Staff | to log notes describing repairs made and spare parts replaced                   | hardware inventory and repair history are preserved | FR-003 |
| HMT-19 | 3.5 On-Hold Pending Parts Flag  | Maintenance Staff | to mark a ticket as 'On-Hold: Awaiting Spare Parts' with expected arrival dates | SLA escalation timers are fairly paused             | FR-003 |

**Acceptance criteria were taken from Lab 1.** FR-003: Staff member updates status from Assigned -> In Progress -> Resolved; system updates database within 1 second and appends an immutable timestamped log entry to the ticket history.

### 3. Jira Backlog – Epic, User Stories and Story Points

The screenshot shows Epic 3 with all 5 User Stories, the priority icon and story points on each row, and the Epic panel reporting 5 work items and an estimate of 21 points.

The screenshot displays the Jira Backlog for the 'Hostel Maintenance Team07' space. The 'HMT board' is selected, and the 'Backlog' tab is active. The backlog shows 5 work items under 'Epic 3 - Maintenance Work Execution & Progress Tracking'. The items are: HMT-11 (3.1 Assigned Work Queue, 5 pts), HMT-16 (3.2 Status Update To In-Progress, 3 pts), HMT-17 (3.3 Status Update To Resolved, 5 pts), HMT-18 (3.4 Parts Used & Work Log Notes, 5 pts), and HMT-19 (3.5 On-Hold Pending Parts Flag, 3 pts). The epic panel on the left shows the total estimate of 21 points for 5 work items.

Figure 1: Jira Backlog – Epic 3 (HMT-14) with stories (5, 3, 5, 5, 3 = 21 pts).

### 4. Backlog Prioritization

User Stories were ordered by business value and dependency. The backlog order in Jira is 3.1, 3.2, 3.3, 3.4, 3.5.

| Key    | Story                            | Priority | Story Points |
|--------|----------------------------------|----------|--------------|
| HMT-15 | 3.1 Assigned Work Queue          | High     | 5            |
| HMT-16 | 3.2 Status Update to In-Progress | High     | 3            |
| HMT-17 | 3.3 Status Update to Resolved    | High     | 5            |
| HMT-18 | 3.4 Parts Used & Work Log Notes  | Medium   | 5            |

|        |                                |        |   |
|--------|--------------------------------|--------|---|
| HMT-19 | 3.5 On-Hold Pending Parts Flag | Medium | 3 |
|--------|--------------------------------|--------|---|

- 3.1, 3.2, and 3.3 (High): Core state machine transitions; without them, tickets remain perpetually in 'Assigned' and resolution cannot be signaled.
- 3.4 (Medium): Enriches the completed record with inventory usage and audit commentary.
- 3.5 (Medium): Exception handler to manage procurement delays without causing unfair SLA penalties.

**Total Story Points (Epic 3): 21 | High: 13 | Medium: 8**

## 5. Story Point Estimation – Planning Poker

Story points were estimated independently by each team member using the Fibonacci scale (1, 2, 3, 5, 8, 13...). The individual estimates were compared, differences were discussed, and the team agreed on the final estimates.

| Story ID     | Story Title                  | Saanvi | Sonia | Kaustubh | Kasissnu | Agreed |
|--------------|------------------------------|--------|-------|----------|----------|--------|
| HMT-15 / 3.1 | Assigned Work Queue          | 5      | 5     | 5        | 5        | 5      |
| HMT-16 / 3.2 | Status Update to In-Progress | 3      | 3     | 3        | 3        | 3      |
| HMT-17 / 3.3 | Status Update to Resolved    | 5      | 5     | 5        | 5        | 5      |
| HMT-18 / 3.4 | Parts Used & Work Log Notes  | 5      | 5     | 5        | 3        | 5      |
| HMT-19 / 3.5 | On-Hold Pending Parts Flag   | 3      | 3     | 2        | 3        | 3      |

## 6. Sprint Planning & Simulation

### Sprint 1 (Duration: 1 week)

Sprint Goal: Deliver core technician task queue and full state machine transitions (In-Progress, Resolved).

| Key    | Story                            | Priority | Pts | Status |
|--------|----------------------------------|----------|-----|--------|
| HMT-15 | 3.1 Assigned Work Queue          | High     | 5   | Done   |
| HMT-16 | 3.2 Status Update to In-Progress | High     | 3   | Done   |
| HMT-17 | 3.3 Status Update to Resolved    | High     | 5   | Done   |

**Sprint 1 Velocity: 13 pts committed | 13 pts completed | 0 pts rolled over**

### Sprint 2 (Duration: 1 week)

Sprint Goal: Support repair notes, spare parts tracking, and on-hold exception states.

| Key    | Story                           | Priority | Pts | Status |
|--------|---------------------------------|----------|-----|--------|
| HMT-18 | 3.4 Parts Used & Work Log Notes | Medium   | 5   | Done   |
| HMT-19 | 3.5 On-Hold Pending Parts Flag  | Medium   | 3   | Done   |

**Sprint 2 Velocity: 8 pts committed | 8 pts completed | 0 pts rolled over**

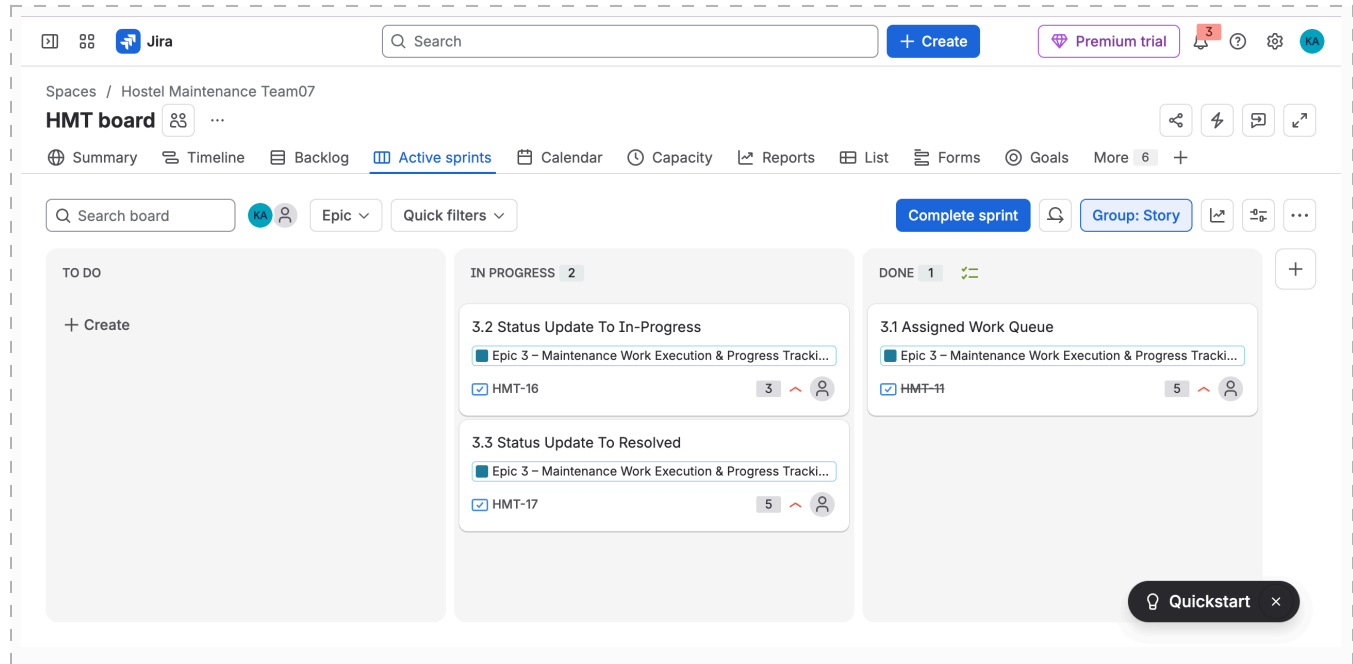
**Jira Sprint Board (Active Sprint view) – Sprint 1**

Figure 2: Sprint 1 Active Sprint board mid-sprint – Story 3.1 in DONE; Stories 3.2 and 3.3 IN PROGRESS.

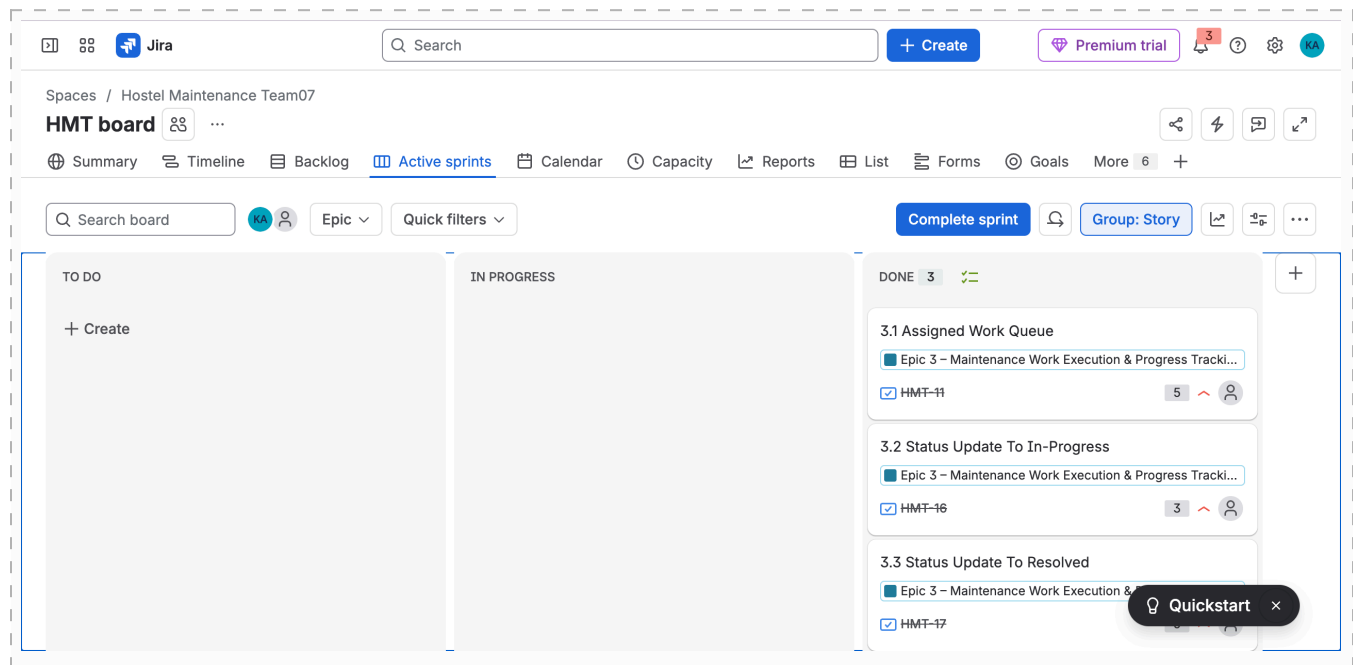


Figure 3: Sprint 1 Active Sprint board – all stories (13 pts) in DONE, before clicking Complete sprint.

## Sprint 2 outcome

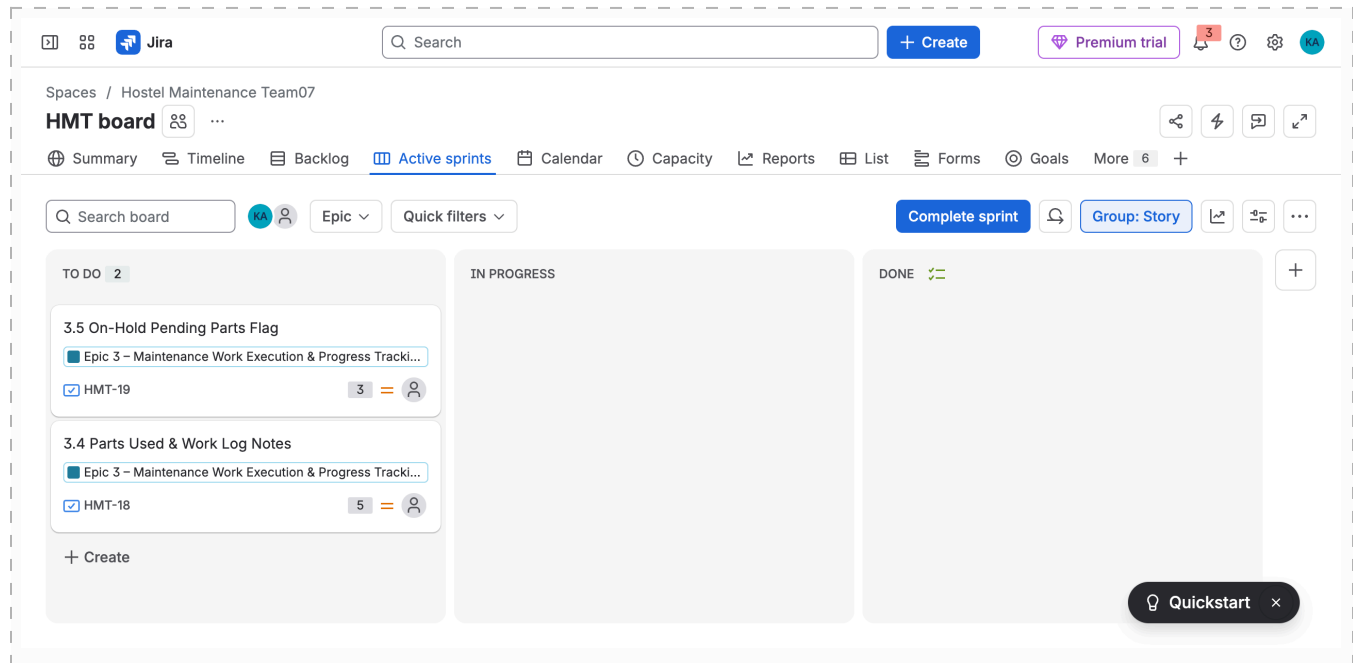


Figure 4: Sprint 2 Active Sprint board – all stories (8 pts) in TO DO.

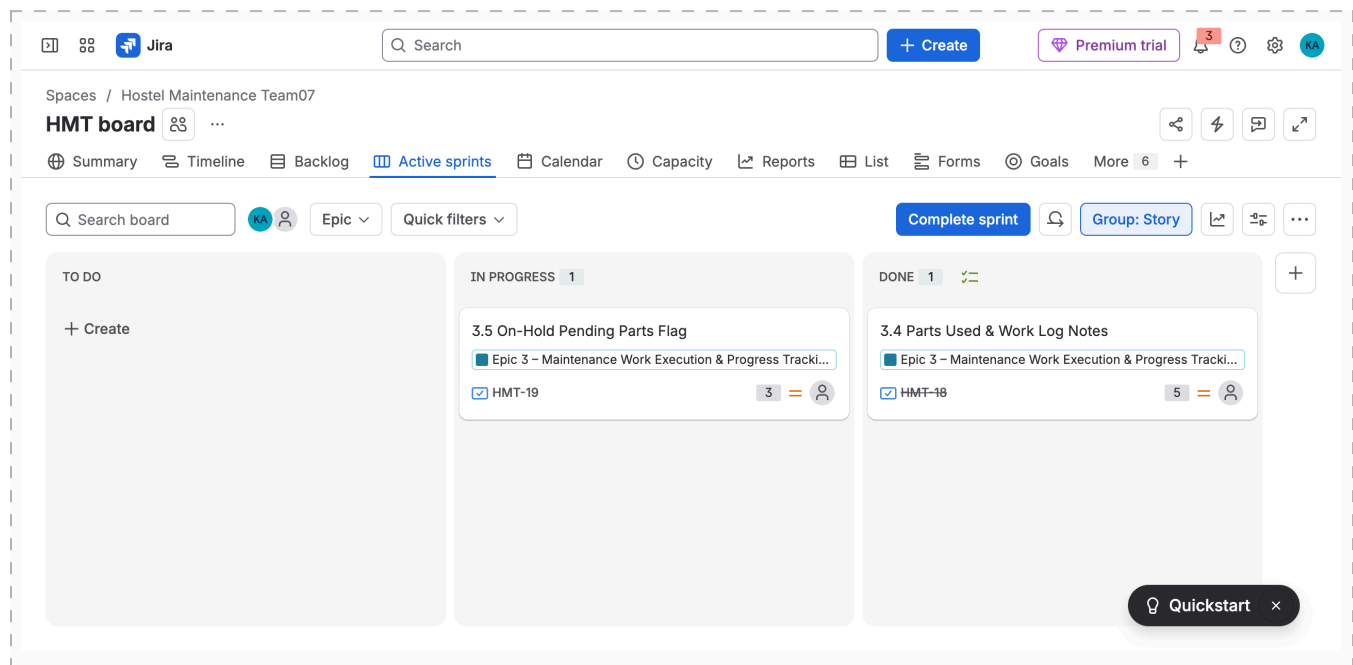


Figure 5: Sprint 2 Active Sprint board mid-sprint – Story 3.4 in DONE; Story 3.5 IN PROGRESS.

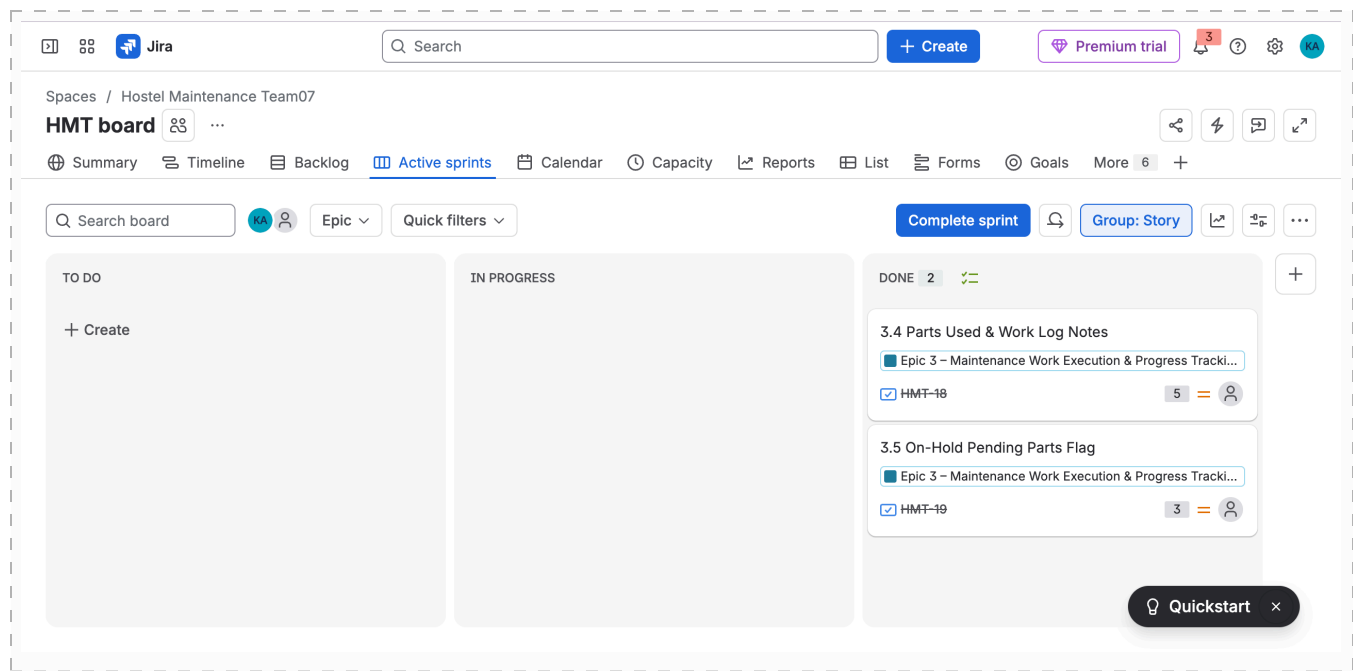


Figure 6: Sprint 2 Active Sprint board – all stories (8 pts) in DONE, before clicking Complete sprint.

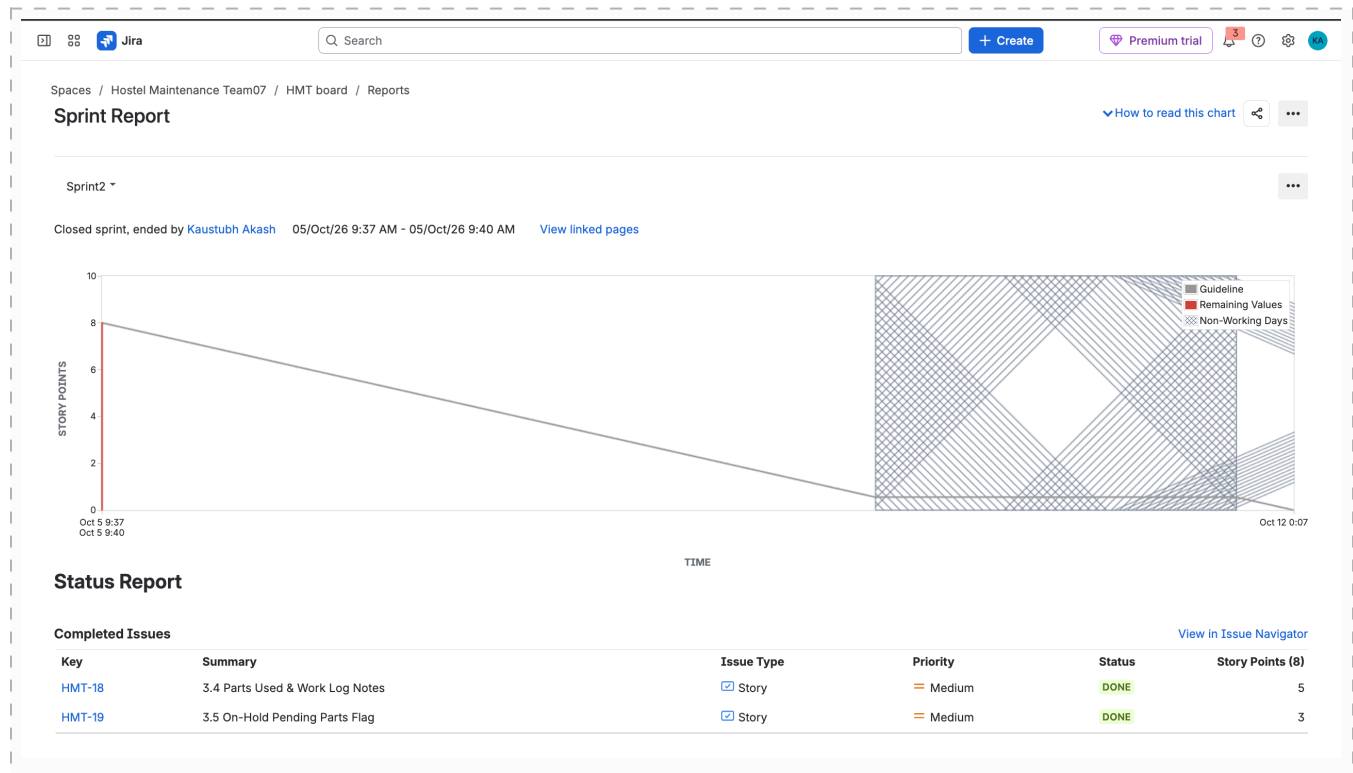


Figure 7: Sprint 2 outcome (Sprint Report, taken after completion) – all stories DONE, 8 story points.

7. Burndown Chart Analysis

The Jira Burndown Chart plots story points remaining (red line) against the ideal guideline (grey line). Each sprint is shown twice: once with non-working days displayed (hatched) and once with them hidden, which makes the guideline easier to read. The event table under each chart lists the exact completion times.

Sprint 1 Burndown (13 pts)

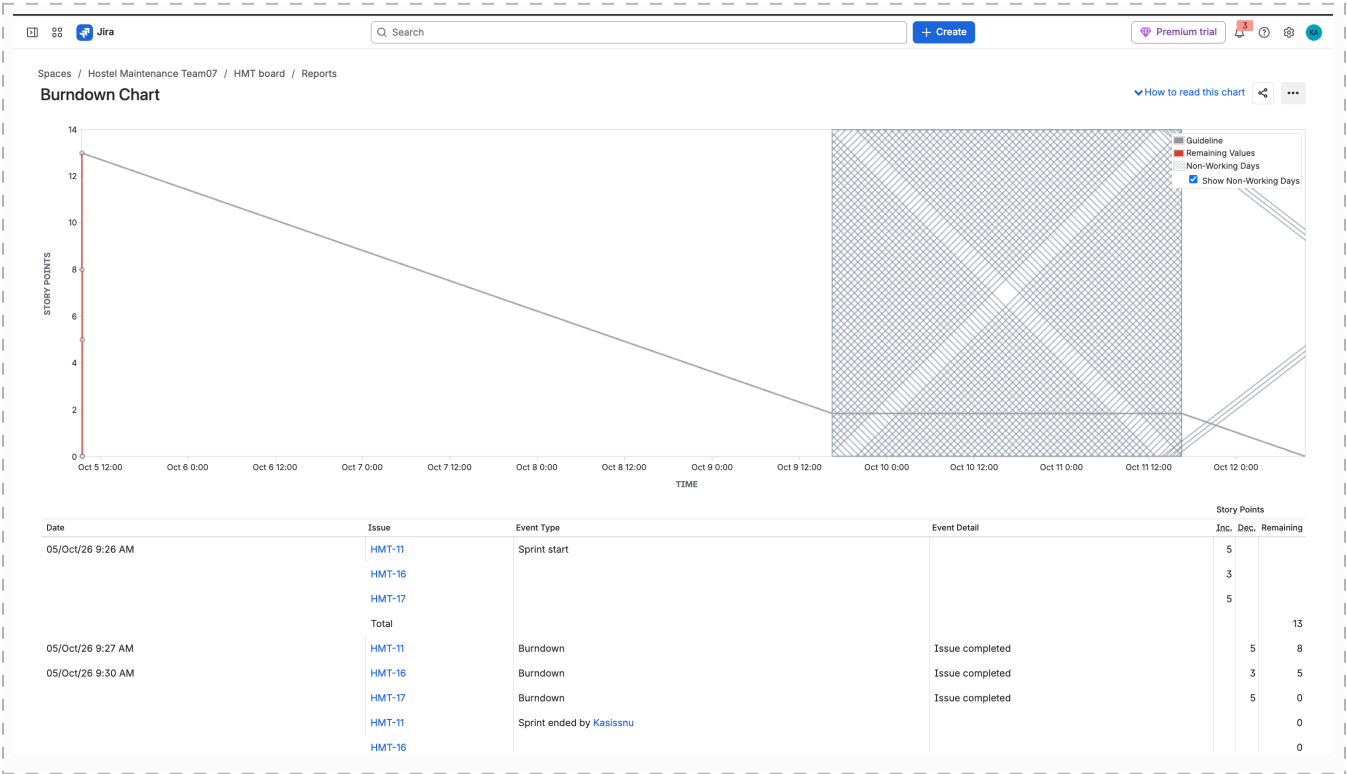


Figure 8: Sprint 1 Burndown, non-working days shown. 13 → 0.

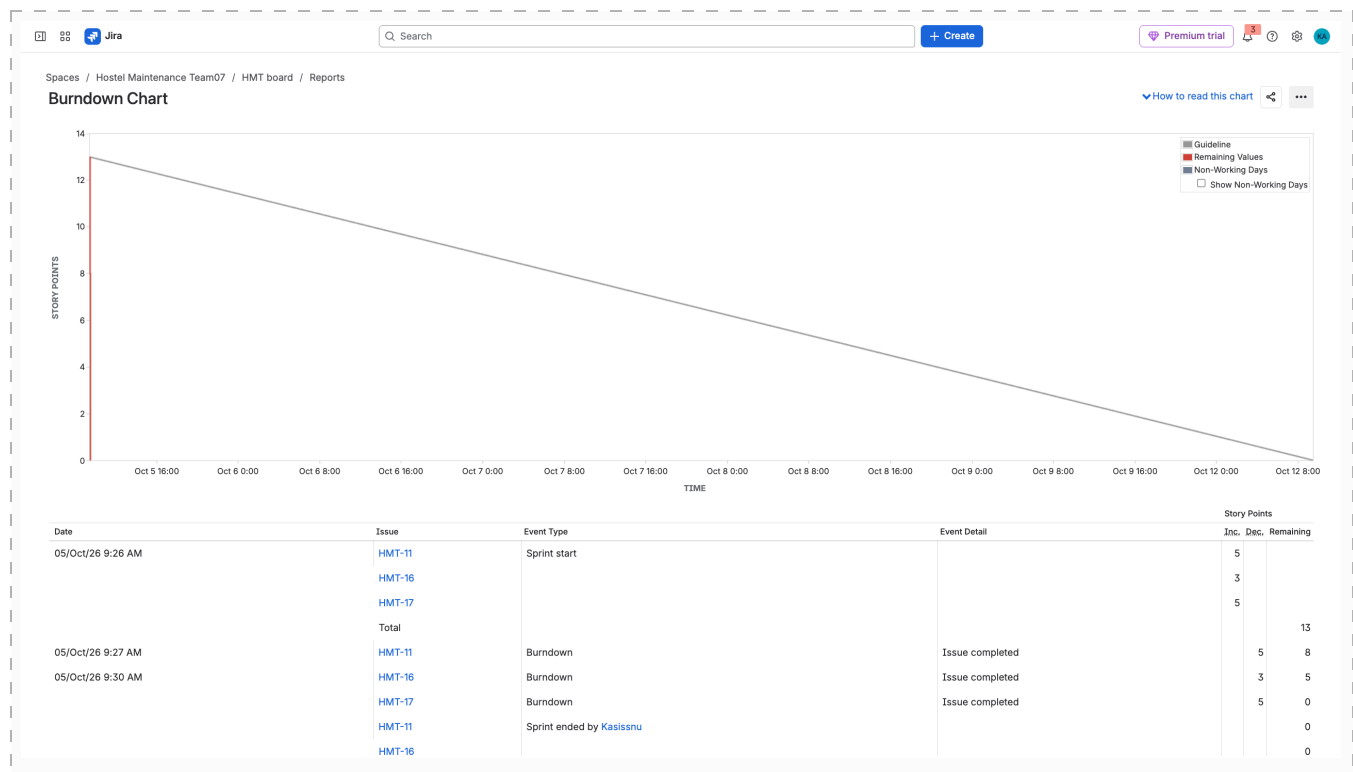


Figure 9: Sprint 1 Burndown, non-working days hidden.

## Sprint 2 Burndown (8 pts)

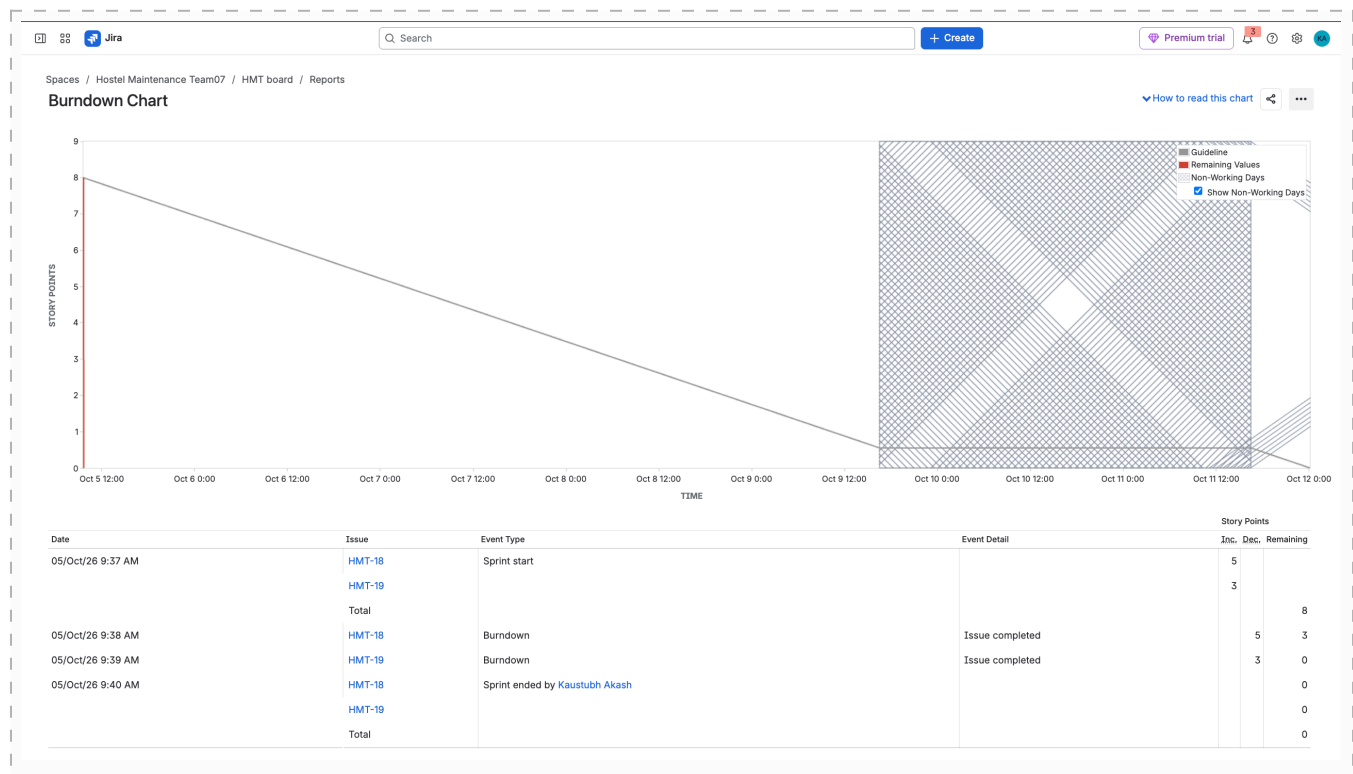


Figure 10: Sprint 2 Burndown, non-working days shown. 8 → 0.

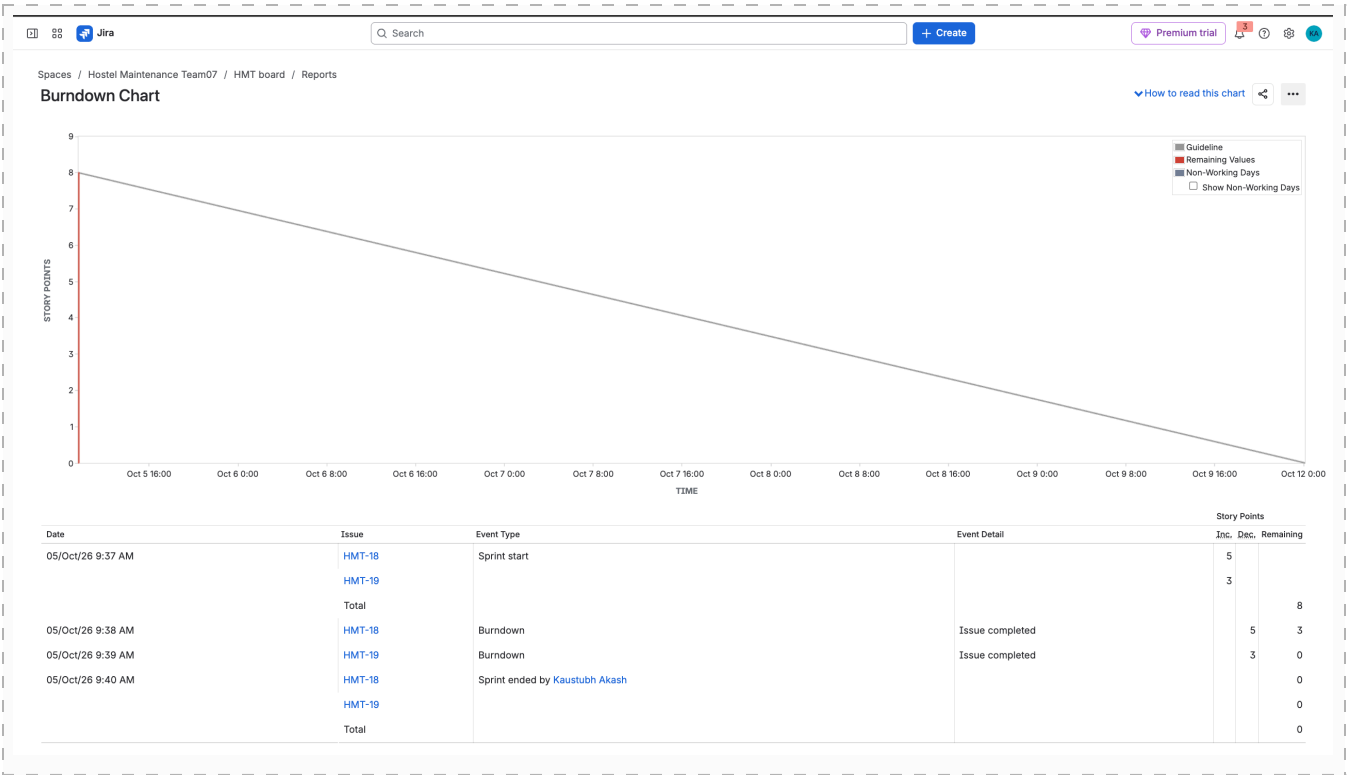


Figure 11: Sprint 2 Burndown, non-working days hidden.

Summary of the burndown data (from the Jira event tables)

| Sprint   | Start (points) | Completion events                                                         | End (points) |
|----------|----------------|---------------------------------------------------------------------------|--------------|
| Sprint 1 | 6:40 PM – 13   | 6:43 PM HMT-15 (–5) → 8; 6:45 PM HMT-16 (–3) → 5; 6:48 PM HMT-17 (–5) → 0 | 0            |
| Sprint 2 | 7:00 PM – 8    | 7:03 PM HMT-18 (–5) → 3; 7:05 PM HMT-19 (–3) → 0                          | 0            |

8. Reflection Questions

1. Did your estimations reflect the actual effort?

Yes. Sizing properly differentiated state-machine concurrency from text updates: Status Update to Resolved (5 pts) requires audit timestamp recording, validation that work was previously In Progress, and triggering resident review prompts. Simpler status transitions (In Progress, 3 pts) were easier to execute.

2. Was your backlog well-prioritized?

Yes. Fundamental status lifecycles were scheduled in Sprint 1 so the end-to-end ticketing cycle could function. Inventory and parts notes were scheduled in Sprint 2 as secondary auditing features.

3. How did your simulated sprint align with your plan?

The simulated sprint aligned exactly with the commitment: 13 points were burned down in Sprint 1 and 8 points in Sprint 2. Zero stories carried over.

4. What insights did the burndown chart give about your team's capacity?

Sprint 1 reached 13 points, which felt like the optimal velocity ceiling for this epic. Sprint 2 burned down 8 points rapidly, indicating that the team had spare capacity in Sprint 2 that could accommodate additional reporting or analytics stories.